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Research Article

Effects of laser oscillation on metal mixing, microstructure, and mechanical property of Aluminum–Copper welds

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ABSTRACT

Laser welding of dissimilar metals is important in many industrial applications. However, as dissimilar metals get mixed during the melting process, intermetallic compounds are often formed in the welds which can significantly undermine the electrical and mechanical properties of the welds. This poses a critical challenge to the widespread utilization of this welding technique. Compared with conventional line-scan laser welding, oscillating laser welding offers additional processing parameters to control the welding process. Although some work has been reported on oscillating laser welding of dissimilar metals, a mechanistic understanding of this process was still missing. The research objective of this work was to reveal the physical mechanisms and evaluate their relative significance to the fluid flow, metal mixing, and microstructure evolution in the molten pool in oscillating laser welding of dissimilar metals. A combination of experiments and simulations was leveraged to achieve the objective. Four fluid flows have been found to determine the metal mixing in the molten pool, and their dependences on the laser oscillating parameters were discussed. In addition, the thermo-solutal conditions of the molten pool solidification were quantified as functions of the laser oscillating parameters, and the effects of the thermo-solutal conditions on the final weld microstructures were analyzed.

1. Introduction

Compared with conventional welding methods of joining dissimilar metals, such as arc-based welding, resistance spot welding, and ultrasonic welding, laser welding has the advantage of high speed and non-contact. It has, therefore, drawn attention from various industrial fields, such as automotive, aerospace, and electronics [1]. However, a common challenge for all fusion-based welding methods, including laser welding, for dissimilar metal joining is the formation of detrimental intermetallic compounds (IMCs) in the welds as dissimilar metals are melted and mixed in the molten pool. As IMCs are generally brittle materials with high electrical resistances, they can significantly undermine the mechanical and electrical performances of the welds [2–4].

Prior work has focused on investigating the relationship of metal mixing, microstructure, and mechanical performance in welding dissimilar metals using line-scan lasers. Fortunato et al. [5] compared the metal mixing results in the fusion zone for laser keyhole welding of Aluminum (Al)-on-Copper (Cu) and Cu-on-Al configurations. Vortex mixing patterns were observed in the Al-rich region for both stack-ups. Shin et al. [6] studied pulsed laser welding of Al ribbon to Cu sheet for the electrical interconnections. They identified the formation of several IMCs in the welds, such as Al_4Cu_9, Al_2Cu, and AlCu. These IMCs were the cause of cracks and voids in the welds. Tomashchuk et al. [7] studied the metal mixing of laser keyhole welding of butt joint steel and Cu by using a simplified two-dimensional model. Since the model used a predefined circular keyhole, the effects of dynamic keyhole fluctuation on the fluid flow and metal mixing were not simulated. Wu et al. [8] developed a three-dimensional (3D) model for laser keyhole welding of lap joint stainless steel and titanium. Since the model only considered diffusion but neglected advection, only a diffusion layer could be observed at the interface between the two metals. Recently, Huang et al. [9] conducted a comprehensive study on the metal mixing process in laser keyhole welding of Al and Cu in a lap joint configuration by using a combination of experiment and simulation. They found that the recoil pressure contributes to an upward flow that pushes Cu to migrate from the bottom of the molten pool to the top, and the Marangoni force generates one backward flow and two side vortices that facilitate the mixing of Al and Cu in the upper region of the molten pool. Such work has revealed the

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underlying physics that governs the mixing of dissimilar metals and IMCs formation during welding. However, for line-scan laser welding, the adjustments in laser power and welding speed can change the magnitude of the flow velocity, but the fluid flow pattern does not change [9]. With this fixed flow pattern, the mixing of dissimilar metals cannot be effectively controlled.

In recent years, oscillating laser welding has drawn more attention and shown new promises for dissimilar metal welding. By tunning laser oscillating parameters (e.g., oscillating pattern, amplitude, and frequency) as the additional degree of freedom, it offers more control over the metal mixing and microstructure of the welds and an establishment of the process-microstructure-property relationship for laser welding of dissimilar metals. It was found that the oscillating laser beam could greatly improve the uniformity of the dissimilar metal mixing in the welds [10,11]. For fully miscible nickel and Cu, Jiang et al. [11] found that a thorough mixing resulted in solid solution strengthening in the welds, which in turn increased the tensile strength of the welds by 20%. For the non-miscible Al and Cu welds, Solchenbach et al. [12–14] found that the oscillating laser welding could reduce the IMCs layer thickness at the fusion zone boundary and hence improve the strength of the welds. For some other non-miscible metal welds (steel-Al, steel-Cu, and Al-magnesium), it was reported that the oscillating laser beam could generate a chaotic fluid flow in the molten pool, which possibly scattered the precipitated IMCs or increased the needle-shaped IMCs in the welds. These IMCs morphologies could strengthen the welds [15–17]. Specifically, Shah et al. [15,16] studied oscillating laser welding of lap joint Al and magnesium alloy with a nickel foil interlayer in between the two metals. They found that oscillating laser welding could effectively reduce cracks and improve the mechanical performance of the welds. They speculated that by improving the dissimilar metals mixing via oscillating laser welding, the formation of continuity-shaped IMCs could be avoided. Instead, IMCs could be dispersed uniformly in the fusion zone due to the keyhole stirring, which could strengthen the welds. Yan et al. [17] studied the microstructure and mechanical properties of steel-Cu welds by using oscillating laser welding. Their results showed that compared with conventional line-scan welding, the tensile strength and elongation of the welds increased 110% and 153% by using oscillating laser welding. They speculated that a vortex flow was generated in the molten pool due to the oscillating laser beam, thereby during the molten pool solidification process, precipitated phases in the welds were scattered in the welds due to this flow, which improved weld quality.

While these experimental studies have revealed the effects of the oscillating laser beam on the process-microstructure-property relationship, the fundamental physics was not clearly understood. Since it is technically challenging to perform in-situ experiments to observe metal mixing dynamics and microstructure evolution, computational fluid dynamics (CFD) simulations and computational materials models can be employed to understand these important matters. To simulate molten pool dynamics, CFD was used for oscillating laser welding of one- or two-layer configuration. For studies on the fluid flow in the molten pool, Zhang et al. [18] found that an oscillating laser beam stirred the molten pool, in which a vortex fluid flow was formed. Other researchers studied the keyhole and molten pool dynamics and the formation of pores and spatter. For instance, Chen et al. [19,20] found that the molten pool shape changed from an elongated droplet to an ellipse with an increasing oscillating frequency. Ke et al. [21] found that both circular-shaped and infinity-shaped oscillating patterns could generate a wider and more stable keyhole and a larger and shallower molten pool that prevented keyhole collapse and therefore inhibited pore formation. Wan et al. [22] investigated oscillating laser welding of thin-gage zinc-coated steel, and found that when the laser oscillating frequency was larger than the keyhole fluctuation frequency, a stable keyhole and zinc outgassing could be achieved, and hence the spatter was reduced. To the best knowledge of the authors, there has been no modeling work that investigates the effect of laser oscillation on the mixing of dissimilar metals during welding.

For computational materials modeling, abundant work has been carried out for laser welding and other similar processes of single metals. These modeling studies can be divided into two categories based on the specific microstructure problems that they targeted: phase composition models and microstructure morphology models. Phase composition models used various algorithms (e.g., Johnson-Mehl-Avrami and Koistinen-Marburger kinetics models, and continuous cooling transformation curve) to predict the phase transformation of martensite, bainite, ferrite, and pearlite in the fusion zone in laser welding of steels. Kubiak et al. [23–26] conducted a series of modeling studies to simulate the volume percentage of martensite, bainite, ferrite, and pearlite in the fusion zone after laser processing of steels. Their model was based on the classic Johnson-Mehl-Avrami and Koistinen-Marburger kinetics models. Morawiec et al. [27] studied the phase transformation simulation of laser-welded steel by using the combination of Sysweld and JMatPro software. They found that martensite concentration was highest in the fusion zone and was decreased in the heat-affected zone. Murthy et al. [28] built a phase transformation model based on the Johnson-Mehl-Avrami-Kolmogorov equation for the laser welding of low alloy steels. The distributions of austenite, bainite, and martensite in the fusion zone were calculated from the model. The microstructure morphology models include a series of new modeling techniques (e.g., cellular automata, Monte Carlo, and phase field) that can simulate the grain or dendrite morphology development in laser welding of single metals (e.g., steels, Al alloys, and Inconel alloys). For grain-level microstructure morphology models, Wei et al. [29–31] conducted an extensive amount of work to investigate the evolution of 3D grain structure during laser keyhole welding of different metals (e.g., Cu, Al alloys, and IN718) using the Monte Carlo method. For dendrite-level microstructure morphology models, Geng et al. [32] developed a cellular automata model to study the dendritic growth at the fusion zone boundary in laser welding. Xiong et al. [33] used the phase field to simulate the columnar-to-equiaxed transition in the entire molten pool of laser welding of Al–Cu alloy. While abundant simulations have been performed for laser welding of single metals, few researchers have investigated the microstructure evolution in laser welding of dissimilar metals, which can be more complex. Specifically, the molten pool solidification in the welding process is affected not only by the non-uniform thermal history, but also by the non-uniform chemical composition. The spatial variations of the thermal history and chemical composition (referred to as the thermo-solutal condition hereafter) across the entire molten pool can generate very different microstructures at different locations within the welds.

Despite all the prior investigations, a mechanistic understanding of the process-microstructure-property relationship is still missing for the oscillating laser welding of dissimilar metals. Specifically, few works can be found to explicitly discuss the fundamental physics that drive fluid flow, heat transfer, metal mixing, and the microstructure evolution during molten pool solidification. The objective of this work is to reveal the physical mechanisms and evaluate their relative significance to the heat/solute transfer and molten pool solidification in oscillating laser welding of dissimilar metals. The knowledge generated from this work can also provide mechanistic guidelines for other fusion-based processes involving multi-materials, such as arc/electron-beam welding of dissimilar metals [34,35] and laser/electron beam additive manufacturing of functionally graded materials [36,37].

2. Methodology

To achieve the research objective, a combination of experiment and simulation was used in this work. Parametric experiments were first conducted to identify the effects of different laser oscillating parameters on the metal mixing, microstructure, and mechanical performance of the welds. A numerical framework was then developed to capture the process-microstructure relationship: a CFD model was developed to simulate the fluid flow and metal mixing during welding, and a computational materials model was developed to predict the phase distributions in the welds based on the predicted distribution of thermo-solutal conditions.

2.1. Experiments

Fig. 1 shows the schematic of a lap joint configuration for the oscillating laser welding of dissimilar metals. Al and Cu were selected due to their applications in electric vehicles. A commercially pure Al (1100) sheet of 0.2 mm thickness was placed on top of a commercially pure Cu (110) sheet of 0.5 mm thickness. The fiber laser had a maximum power output of 2000 W at a wavelength of 1064 nm. The laser spot size was 115 μm at the focal plane on the top surface of the Al sheet. Since the laser absorption rate of Al is about 3.5 times the laser absorption rate of Cu at the given laser wavelength at room temperature, the Al-on-Cu stack-up can facilitate the formation of the molten pool and keyhole for the welding process. A circular oscillating path was selected for the laser movement to perform the welding process. It was considered that the circular oscillating path could alter the metal mixing due to the keyhole-stirring effect captured by the high-speed camera in laser welding of a single metal [38].

To investigate the effects of different oscillating parameters on the metal mixing, microstructure, and mechanical performance of the welds, sixteen cases with different oscillating amplitudes and frequencies were studied, as shown in Table 1. This parametric matrix was selected to ensure a variety of metal mixing statuses in the fusion zone among the sixteen cases for analysis purposes. The oscillating amplitude (denoted as A in Fig. 1) is the maximum deviation of the laser oscillating path in the transverse direction. Oscillating frequency (denoted as f in Fig. 1) represents the number of circular patterns that the laser passes in every second. Laser power was fixed at 630 W and the laser forward speed V_f was fixed at 0.08 m/s to ensure the molten pool penetration of the top Al sheet and non-penetration of the bottom Cu sheet for all cases. Since the welding length was fixed at 40 mm for all cases, the welding time was 0.5 s for all cases to ensure the same welding efficiency. The laser oscillating speed V_o can be calculated by the following equation.

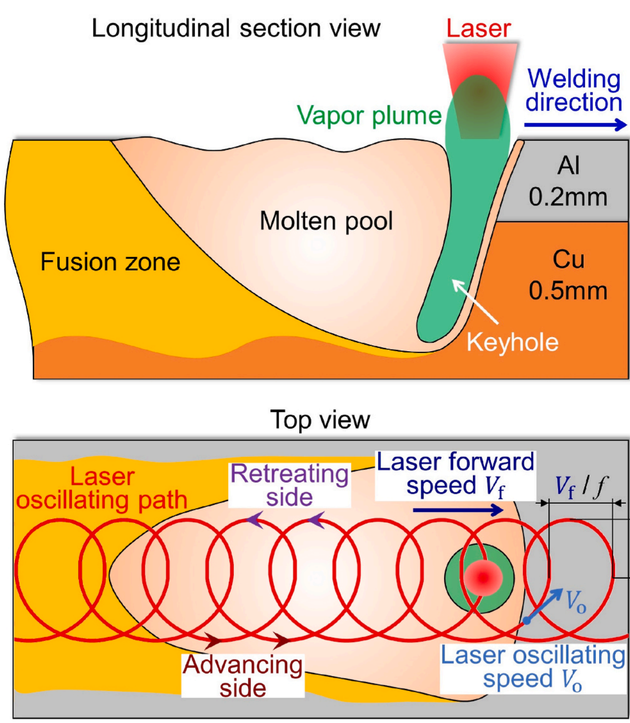

Fig. 1. Schematic of oscillating laser welding for Al–Cu lap joints.

Table 1
Experimental parameters.

f / A	300 Hz	600 Hz	900 Hz	1200 Hz
0.05 mm	#1	#2	#3	#4
0.10 mm	#5	#6	#7	#8
0.15 mm	#9	#10	#11	#12
0.20 mm	#13	#14	#15	#16

$$V_o = 2\pi A f \quad (1)$$

After welding, the welds were cut along their cross-sections and the cut surfaces were polished. The Cu concentration field in the cross-sections was characterized by energy-dispersive X-ray spectroscopy (EDS) element mapping. Scanning electron microscope (SEM) with backscattered electrons and EDS analysis were used to characterize the microstructures and phases of the welds. Mechanical performance testing was performed to identify the peak loads of the welds.

2.2. CFD model

A 3D CFD model was developed using the Flow-3D® software. This model takes the welding parameters (laser power, laser spot size, laser oscillating path and speed, etc.) and material attributes (chemical composition, physical properties, feedstock geometry, etc.) as inputs, and predicts the molten pool and keyhole shapes as well as temporal and spatial variations of temperature, fluid flow, and chemical distribution in the molten pool.

The computational domain of the model is shown in Fig. 2, which only represents a small subset of the welding workpiece. The coupon thickness for the material and laser parameters were the same for the model as in the experiment. The calculation domain was divided into two mesh blocks - a fine mesh block with a uniform mesh size of 10 μm and a coarse mesh block with a uniform mesh size of 50 μm. The molten pool was bounded within the fine mesh block where major physics (i.e., laser-material interaction, keyhole evolution, fluid flow, and metal mixing) were simulated. The coarse mesh block was designed for the heat conduction process in metals. A uniform mesh of 10 μm was chosen for the fine mesh block after a mesh convergence study. Different boundary conditions were imposed on the simulation domain to approximate the conditions in the experiment. Since the top and the bottom parts of the calculation domain are air, the boundary conditions are set to be the constant pressure (atmospheric pressure) boundary condition. Since the four sides of the calculation domain are base metals, the boundary conditions are set to be the outflow boundary condition. The timestep size was set according to the Courant-Friedrichs-Lewy convergence condition. The simulations were run on a 24-core machine with an Intel Xenon CPU 3.00 GHz and 384 GB RAM.

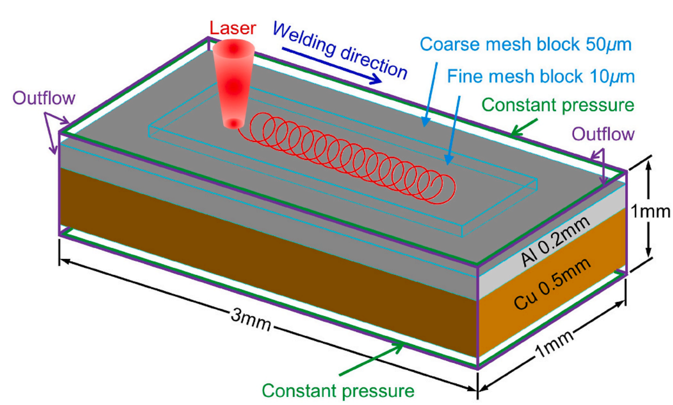

Fig. 2. Model setup and computational domain.

In this model, the velocity field, temperature field, and Cu concentration field are all calculated for the solid and liquid phases with the following governing equations.

$$\frac{\partial}{\partial t}(\rho)+\nabla\left(\rho\vec{V}\right)=0 \quad (2)$$

$$\frac{\partial}{\partial t}\left(\rho\vec{V}\right)+\nabla\left(\rho\vec{V}\vec{V}\right)=\nabla\left(\mu\nabla\vec{V}\right)-\nabla P-\frac{\mu}{K_i}\vec{V}+\rho g \quad (3)$$

$$\frac{\partial}{\partial t}(\rho h)+\nabla\left(\rho\vec{V}h\right)=\nabla(k\nabla T) \quad (4)$$

$$\frac{\partial}{\partial t}(\rho Y)+\nabla\left(\rho\vec{V}Y\right)=\nabla(D\rho\nabla Y) \quad (5)$$

In the mass conservation equation (Eq. (2)), ρ is the density of the fluid, and $\vec{V}$ is the fluid velocity. In the momentum conservation equation (Eq. (3)), μ is the fluid dynamic viscosity, P is the pressure, K_i is the isotropic permeability expressed by the Kozeny-Carman equation, and g is gravity. The last two terms on the right-hand side of Eq. (3) are the source terms that represent the damping force for the liquid phase and the gravitational force. In the energy conservation equation (Eq. (4)), T is the temperature, h is the material enthalpy, and k is the thermal conductivity. In the species conservation equation (Eq. (5)), Y is the Cu concentration in weight percentage, and D is the mass diffusivity. The thermal properties of Al and Cu used in this work are temperature-dependent and are listed in Table 2 [39–44].

The Volume-Of-Fluid (VOF) method [45–47] is employed to track the evolution of the interface (including the molten pool surface and keyhole wall). Eq. (6) shows the governing equation of VOF.

$$\frac{\partial f}{\partial t}+\nabla\left(f\bullet\vec{V}\right)=0 \quad (6)$$

Here f is the fraction of liquid volume over cell volume and $\vec{V}$ is the fluid velocity obtained from the solution from Eq. (2) and Eq. (3). The VOF method defines $f=1$ in the cells of liquid and solid, $f=0$ in the cells of gas, and $1>f>0$ in the cells on the interface. By solving Eq. (6), the distribution of f will be updated, and the interface appears in the cells with $1>f>0$.

There are three major forces on the interface. The first is the recoil pressure caused by metal evaporation, which can be calculated with Eq. (7) [48].

$$P_{\text{recoil}}=\alpha P_{\text{v}}\bullet\exp\left[\frac{\Delta H_{\text{v}}}{(\gamma_{\text{v}}-1)c_{\text{v}}T_{\text{v}}}\left(1-\frac{T_{\text{v}}}{T}\right)\right] \quad (7)$$

Here α is the evaporation coefficient, P_{v} is evaporation pressure, ΔH_{v} is the latent heat of evaporation, γ_{v} is the heat capacity ratio of metal vapor, c_{v} is the specific heat of metal vapor, T_{v} is the evaporation temperature, and T is the temperature at the keyhole wall. The direction of the recoil pressure is along the normal direction of the keyhole wall and points toward the liquid side. The second force is the Laplace pressure P_{L}, which is caused by the surface tension on a curved interface as calculated by Eq. (8).

$$P_{\text{L}}=\gamma\kappa \quad (8)$$

Here γ and κ are the surface tension and interface curvature, respectively. γ is a function of temperature T and Cu concentration Y. The third force is the Marangoni force caused by the gradient of surface tension on the interface. The spatial variations of temperature and Cu concentration on the interface cause a gradient of surface tension. The surface tension gradient caused by the temperature gradient is the thermo-capillary stress, which has been included in the majority of laser keyhole welding models. The surface tension gradient caused by the Cu concentration gradient is the soluto-capillary stress, which has barely been mentioned in the existing literature. The thermo-capillary stress and the soluto-capillary stress altogether form the Marangoni stress σ_{M} which can be calculated by Eq. (9).

$$\sigma_{\text{M}}=\nabla_{\text{s}}\gamma \quad (9)$$

Here ∇_{s} calculates the gradient along the tangent direction of the interface. The γ is a function of temperature T and Cu concentration Y.

The laser experiences multiple reflections in laser keyhole welding, so the ray-tracing model is used to model this phenomenon [49–51]. This method divides the laser beam into multiple rays. For each ray, the multiple reflection route is tracked by the law of reflection, and the laser power absorbed by the keyhole is calculated by the Fresnel equation. In this work, the built-in ray-tracing function in Flow-3D® is used to predict the laser absorption in the keyhole. The absorption rates of Al and Cu are calculated as temperature-dependent [52].

2.3. Computational materials model

The module of Scheil solidification with back diffusion in the Thermo-Calc® software was used in this work. This model used the predicted thermo-solutal condition for a given point (obtained from the CFD model) as the inputs, and predicted the solidification path and fraction of different phases in the final microstructure. The thermodynamic and mobility data needed for the calculation was directly provided by the Thermo-Calc® database.

The model assumes an infinitely fast diffusion in the liquid phase and a limited diffusion in the solid phase [53–57]. The Scheil equation is written as Eq. (10).

$$C_S=kC_o[1-(1-\beta K)f_S]^{(K-1)/(1-\beta K)} \quad (10)$$

Here C_S is the concentration of solute in the solid, C_o is the initial concentration of the liquid, which is set as the Cu concentration obtained from the CFD model, f_S is the fraction of solid, $K=C_S/C_L$ is the partition coefficient, C_L is the concentration of solute in the liquid, and β is the back diffusion parameter, which is defined in Eq. (11).

$$\beta=2\alpha=\frac{8D_St_f}{\lambda^2} \quad (11)$$

Here α is the mass Fourier number, D_S is the element mass diffusivity in solid, λ is the secondary dendrite arms spacing, and $t_f=(T_L-T_s)/C_R$ is the local time of solidification, T_L is the liquidus temperature, T_s is the solidus temperature, and C_R is the cooling rate which is obtained from the CFD model and is the other initial condition.

Table 2
Thermal properties of Al and Cu as functions of temperature T (K) at solid (s) and liquid (l) states.

Property	Al	Cu
ρ (kg/m^3)	$-4.99\times10^{-4}T^2+0.322T+2648$ (s) $-0.299T+2670$ (l)	8960 (s) $-0.807T+9072$ (l)
k (W/m•K)	$-0.067T+248$ (s) $-2.099\times10^{-5}T^2+7.892\times10^{-2}T+33.9$ (l)	7.509×10^{-2}T $+418.8$ (s) $4.976\times10^{-2}T+89.7$ (l)
c_p (J/kg•K)	1199 (s) 1127 (l)	385 (s) 572 (l)
L_{fus} (J/kg)	3.65×10^5	2.05×10^5
L_{vap} (J/kg)	1.09×10^7	4.73×10^6
γ (mN/m)	$-0.127T+993$ (l)	$-0.176T+1552$ (l)
μ (mPa•s)	$10^{(-0.7324\,+\,803.49/T)}$ (l)	$10^{(-0.4220\,+\,1393.4/T)}$ (l)
T_{m} (K)	933	1358
T_{v} (K)	2743	2835
D (m^2/s)	$2.0\times10^{-11}T-12.4\times10^{-9}$ (l)	$7.5\times10^{-12}T-7.0\times10^{-9}$ (l)

3. Results and discussion

3.1. Experiments

Six cross-sections were cut from the welds for each case. One representative cross-section is shown for each case in the following analysis due to the similarity of all six cross-sections. The cross-section results for the sixteen cases can be found in Fig. 3.

Al and Cu are not completely mixed over the entire fusion zone. Instead, two regions of different concentration fields are formed in the fusion zone (denoted as Region I and Region II hereafter, as labeled in case #1 in Fig. 3). The boundary between Region I and Region II could be easily identified by the concentration map. Region I is primarily an Al-rich region with some Cu mixed in and Region II is primarily a Cu-rich region with some Al mixed in. Cracks are found in Region I for cases #1 - #8 and no cracks are formed in Region II for all the cases, indicating the possible formation of brittle IMCs in Region I. Therefore, this study is focused on investigating the metal mixing in Region I and studying the effect of metal mixing on the microstructure formation in Region I.

To quantify the Al–Cu mixing in Region I, the average Cu concentration in Region I (denoted as μ_{Cu}^{I}) and the standard deviation of Cu concentration in Region I (denoted as σ_{Cu}^{I}) are calculated from the EDS element mapping results of the six cross-sections for each case, as shown in Fig. 4(a) and (b). The error bars in Fig. 4(a) and (b) stands for the variations of the μ_{Cu}^{I} and σ_{Cu}^{I} results of the six cross-sections for each case. μ_{Cu}^{I} is a metric to quantify the amount of Cu that flows up from the bottom Cu plate to Region I to mix with Al, and σ_{Cu}^{I} is a metric to quantify the uniformity of the Al–Cu mixing in Region I. For the cases with fixed oscillating amplitudes of 0.05 mm (cases #1 - #4, blue curves in Fig. 4(a) and (b)) and 0.10 mm (cases #5 - #8, orange curves in Fig. 4 (a) and (b)), μ_{Cu}^{I} and σ_{Cu}^{I} do not change significantly with the increment of oscillating frequency. For cases with fixed oscillating amplitudes of 0.15 mm (cases #9 - #12, yellow curves in Fig. 4(a) and (b)) and 0.20 mm (cases #13 - #16, purple curves in Fig. 4(a) and (b)), μ_{Cu}^{I} and σ_{Cu}^{I} decrease with the increment of oscillating frequency. A comprehensive analysis of this phenomenon is presented in the following section using a CFD model.

The representative microstructures in Region I for the sixteen cases can be found in Fig. 5. The phases in the microstructure were identified using EDS. From the cases of high oscillating amplitude and frequency to low oscillating amplitude and frequency (i.e., from case #16 to case #1), μ_{Cu}^{I} generally follows an increasing trend, and the microstructure in the welds change from one case to the next. Granular primary α (i.e., Al solid solution) and eutectic dominates in case #16, eutectic microstructure is found in case #15, granular primary θ (i.e., Al_2Cu) and eutectic dominates in cases #14 - #9, and then bulk primary θ prevails in cases #8 - #1.

For the cases with the high oscillating amplitude of 0.20 mm, cases #16, #15, and #14 present cases with μ_{Cu}^{I} being 9.56 at% (blue texts in Fig. 4(a)), 15.91 at% (red texts in Fig. 4(a)), and 22.25 at% (green texts in Fig. 4(a)), respectively. Based on the Al–Cu phase diagram in Fig. 4(c), the μ_{Cu}^{I} of the three cases reside in the hypoeutectic (blue shades), eutectic (red shades), and hypereutectic (green shades) regions, respectively. Therefore, the resultant microstructure for case #16 is a typical hypoeutectic morphology with primary α granules surrounded by the eutectic phase (blue bordered image in Fig. 5). Primary α first precipitate from the molten pool when the local temperature drops below the liquid temperature, and the remaining liquid surrounding the primary α phase turns into the eutectic phase when the local temperature further drops below the eutectic temperature. For case #15, its microstructure is a typical laminar structured eutectic phase which is composed of the eutectic θ and eutectic α phases (red bordered image in Fig. 5). For case #14, its resultant microstructure is a typical hypereutectic morphology with primary θ granules surrounded by the eutectic phase (green bordered image in Fig. 5). The solidification process for this case is similar to case #16, except that primary θ is the first to precipitate from the molten pool. The chemical composition for case #13 also falls into the hypereutectic region, and hence the typical microstructure is similar to that of case #14, except that the eutectic θ phase takes a slightly larger volume percentage.

μ_{Cu}^{I} further increases with the decreasing oscillating amplitude (i.e., from case #12 to case #1). The values fall in the hypereutectic region (yellow curves in Fig. 4(a)) for cases with an oscillating amplitude of 0.15 mm (i.e., case #9 - #12), and the resultant microstructures are primary θ surrounded by the eutectic phase. With an increasing μ_{Cu}^{I} from 21.23 at% to 31.36 at% for a decreasing oscillating frequency from 1200 Hz to 300 Hz (i.e., from case #12 to case #9), an increasing volume of the precipitated primary θ and hence a decreasing volume of eutectic phase are found in the microstructure (as shown in Fig. 5). As μ_{Cu}^{I} keeps increasing with the oscillating amplitude decreased to 0.10 mm and 0.05 mm (i.e., from case #8 to case #1), primary θ grows and starts to form in bulk. Another IMC phase η′ (i.e., AlCu) also forms in cases of oscillating amplitude 0.05 mm (i.e., case #4 - #1).

Mechanical performance testing was performed to identify the peak loads of the welds. The peak load for each of the sixteen cases was evaluated using a lap-shear test as illustrated in Fig. 6(a). The welds

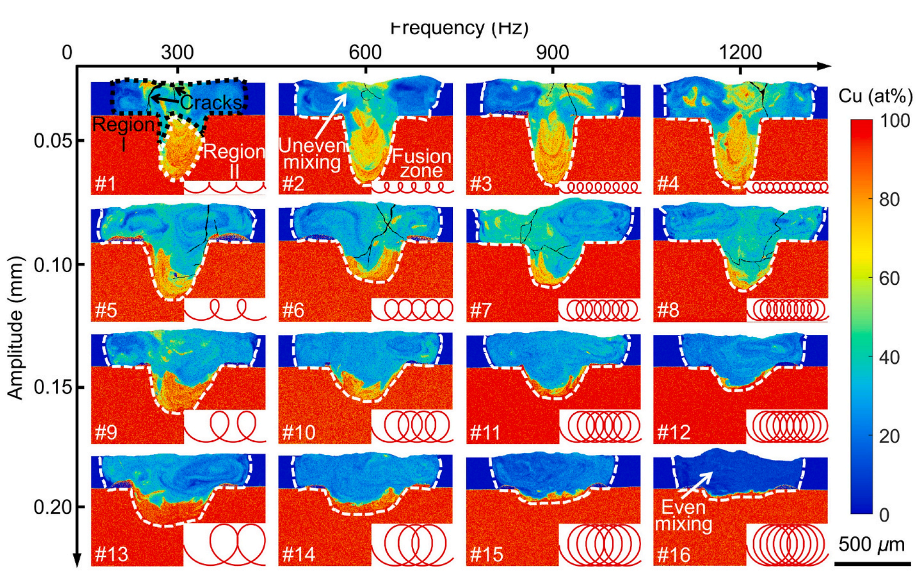

Fig. 3. EDS element mapping results of Cu concentration field (color map), concentration Region I (black dotted curve in case #1) and concentration Region II (white dotted curve in case #1), fusion zone geometry (white dashed curves), cracks (black arrows), mixing patterns (white arrows), and laser oscillating path (red curve) in the cross-section view of the fusion zone for all cases. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

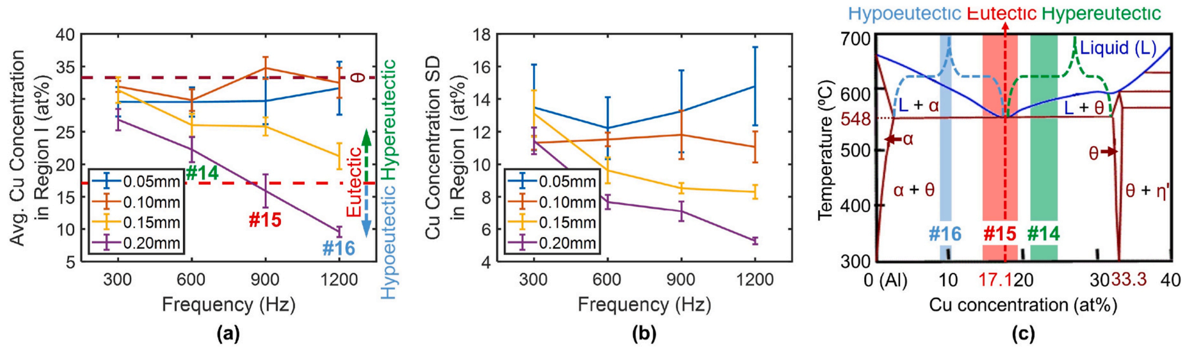

Fig. 4. (a) Average of Cu concentration in Region I and (b) standard deviation of Cu concentration in Region I as functions of oscillating amplitude and frequency, and (c) Al–Cu phase diagram in the eutectic concentration range. The error bars are 1 standard deviation and represent the variations in the value of the y-axis.

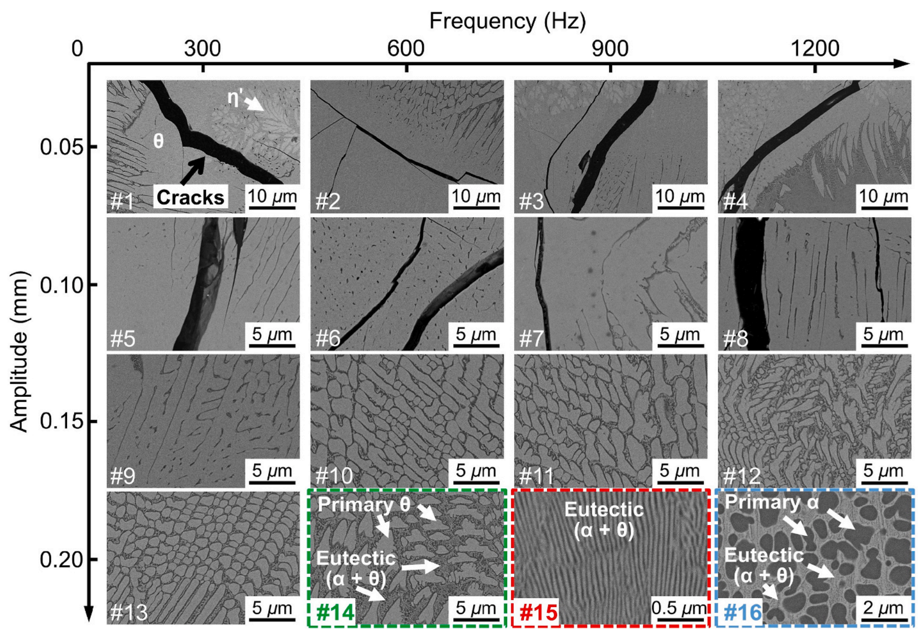

Fig. 5. Experimental results of SEM backscattered electrons imaging of typical microstructure in Region I for all cases, and the phases identified in these microstructures.

were tested at a rate of 1 mm/min at room temperature and the load versus displacement curve including the peak load was recorded, a schematic of which is shown in Fig. 6(b). The results for the sixteen cases are shown in Fig. 6(c). For each case, three specimens were tested. Since the three specimens have the same fracture mode, only one fractured specimen is shown for each case in Fig. 6(c). The average peak load of the three specimens for each case is also shown in Fig. 6(c). The weld peak load is found to be highly dependent on the fusion zone microstructure, especially the morphology of the primary θ phase which has a hardness value of 630HV. In comparison, this value for Al and Cu are 36HV and 75HV, respectively [13].

Welds produced with high oscillating amplitudes (i.e., cases #9 - #16) have higher peak loads than those produced with low oscillating amplitudes (i.e., cases #1 - #8). The fracture mode of the better cases is the base-metal fracture (marked by blue curly brackets in Fig. 6(c)). The typical microstructure of these cases is primary θ or α granules surrounded by the eutectic phase with fine lamellae (as shown in Fig. 5). The continuous network of fine eutectic lamellae with soft α phase and hard θ phase offers a good combination of ductility and strength. The primary phase is fine in size and randomly distributed in the domain, providing abundant grain boundaries to impede the propagation of cracks. The specific peak load varies in different cases depending on the specific type and volume percentage of the primary phase in each case.

The peak loads are lower in low oscillating amplitude cases (i.e., cases #1 - #8), and the fracture usually takes place in the weld zone (marked by red curly brackets in Fig. 6). This is the weld fracture mode. The typical microstructure of these cases is dominated by the brittle θ phase in the bulk shape with few grain boundaries (as shown in Fig. 5). The microstructure is more susceptible to crack formation and propagation, which is responsible for the reduced weld peak load in these cases.

The effects of different laser oscillating parameters on the metal mixing, microstructure, and mechanical performance of the welds have been successfully identified via the experiments. To further reveal the underlying physics of metal mixing and microstructure evolution, a numerical framework will be presented in the following section: a CFD model is included to simulate the fluid flow and metal mixing during welding, and a computational materials model is included to predict the phase formation in the welds.

3.2. CFD model

3.2.1. Model validation

The CFD model has been applied to simulate selected cases (cases #1, #5, #9, and #13 with a fixed oscillating frequency but an increasing oscillating amplitude, and cases #13 - #16 with a fixed oscillating

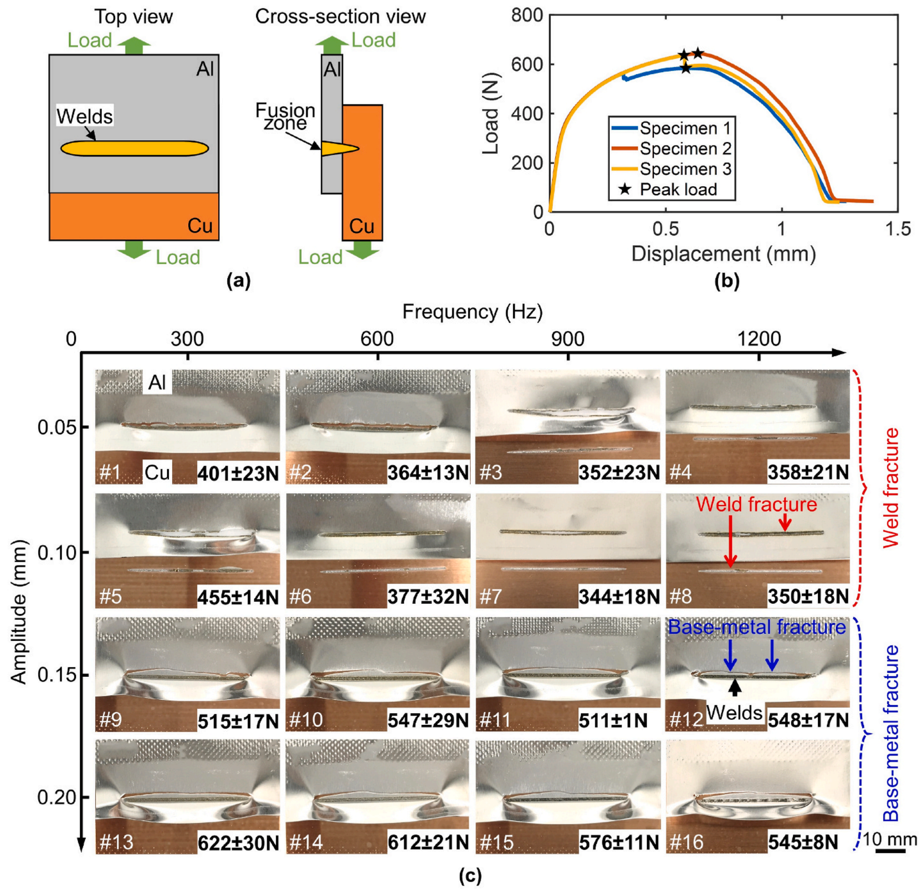

Fig. 6. (a) Schematic of the mechanical performance testing, (b) the load versus displacement curves of the three specimens for case #13, and (c) the typical fractured specimen, fracture mode, and mean peak load for the sixteen cases.

amplitude but an increasing oscillating frequency). Simulation results are validated by comparing their fusion zone geometries and Cu concentration fields against the experimental results. The simulation results of all the selected cases agree reasonably well with the experimental results. Three cases (cases #1, #13, and #16) of the parametric matrix were selected to show the validation results. The three cases are representative cases of the parametric matrix since they cover the upper and lower parametric limits of the matrix, as shown in Fig. 7. The predicted fusion zone geometries and Cu concentration fields are compared with the experimental results side-by-side. For a more quantitative comparison, EDS line scanning comparisons are conducted along a horizontal line (mm') in the middle of Region I and a vertical line (nn') at the center of the fusion zone.

3.2.2. Effects of laser oscillating parameters on metal mixing process

To illustrate the metal mixing process, the simulation results of selected cases are presented in a video in the supplementary material. These two cases have the same oscillating amplitude but different oscillating frequencies, which results in different fluid flow and metal mixing in the molten pool. The keyhole-stirring effect observed from the simulation results matches the same phenomenon captured by the high-speed camera in experiments [38]. The simulation results show that a high oscillating frequency can effectively homogenize metal mixing in the welds due to the high-frequency keyhole-stirring effect.

Supplementary data related to this article can be found online at https://doi.org/10.1016/j.ijmachtools.2023.104020

A further investigation is then focused on the underlying physics regarding metal mixing in the quasi-steady state. A snapshot of the simulation result at 13.25 ms for case #9 is shown in Fig. 8. At this time, the keyhole and molten pool geometries and the metal mixing pattern have all reached quasi-steady states. The simulation result reveals four major fluid flows that affect the metal mixing in the molten pool. The first is the recoil-driven upward flow originating from the keyhole bottom (marked by blue arrows in Fig. 8), which will drive the materials in Region II to flow upward to Region I. The second is the Marangoni-driven flow near the molten pool surface (marked by magenta arrows in Fig. 8), which will drive large-scale vortices (primarily in the planes vertical to the molten pool surface) to facilitate the metal mixing in Region I. These two flows have been discussed extensively in another work [9], where more details can be found. The third is the flow-around-keyhole, which is driven by the recoil pressure generated on the front wall of the fast-moving keyhole (marked by yellow arrows in Fig. 8). It usually drives the fluid in front of the keyhole to move around the keyhole and flow to the rear of the keyhole. Depending upon the conditions associated with the flow-around-keyhole, the fourth flow, which is the small-scale vortices primarily in the horizontal planes deep inside the molten pool (marked by orange arrows in Fig. 8), will likely form. This flow (referred to as keyhole-driven vortices hereafter), if exists, can also effectively facilitate the local metal mixing in Region I.

To better explain the flow-around-keyhole and the keyhole-driven vortices, Fig. 9 shows the two flows on the horizontal section at $z = -0.15$ mm plane for cases #9 and #12. In case #9, the laser oscillating speed is 0.28 m/s and the flow-around-keyhole (marked by yellow arrows in Fig. 9(a)) is as slow as 1.3 m/s. No keyhole-driven vortices can be found. In case #12, the laser oscillating speed is 1.13 m/s and the flow-around-keyhole (marked by the yellow arrows in Fig. 9(b)) is as fast as 2.7 m/s. The keyhole-driven vortices (marked by the orange arrows in Fig. 9(b)) can be found in this scenario. If the flow-around-keyhole can be analyzed as the classical flow-around-a-circular-cylinder problem [58–60], unsteady vortex flow can be formed behind the cylinder if the

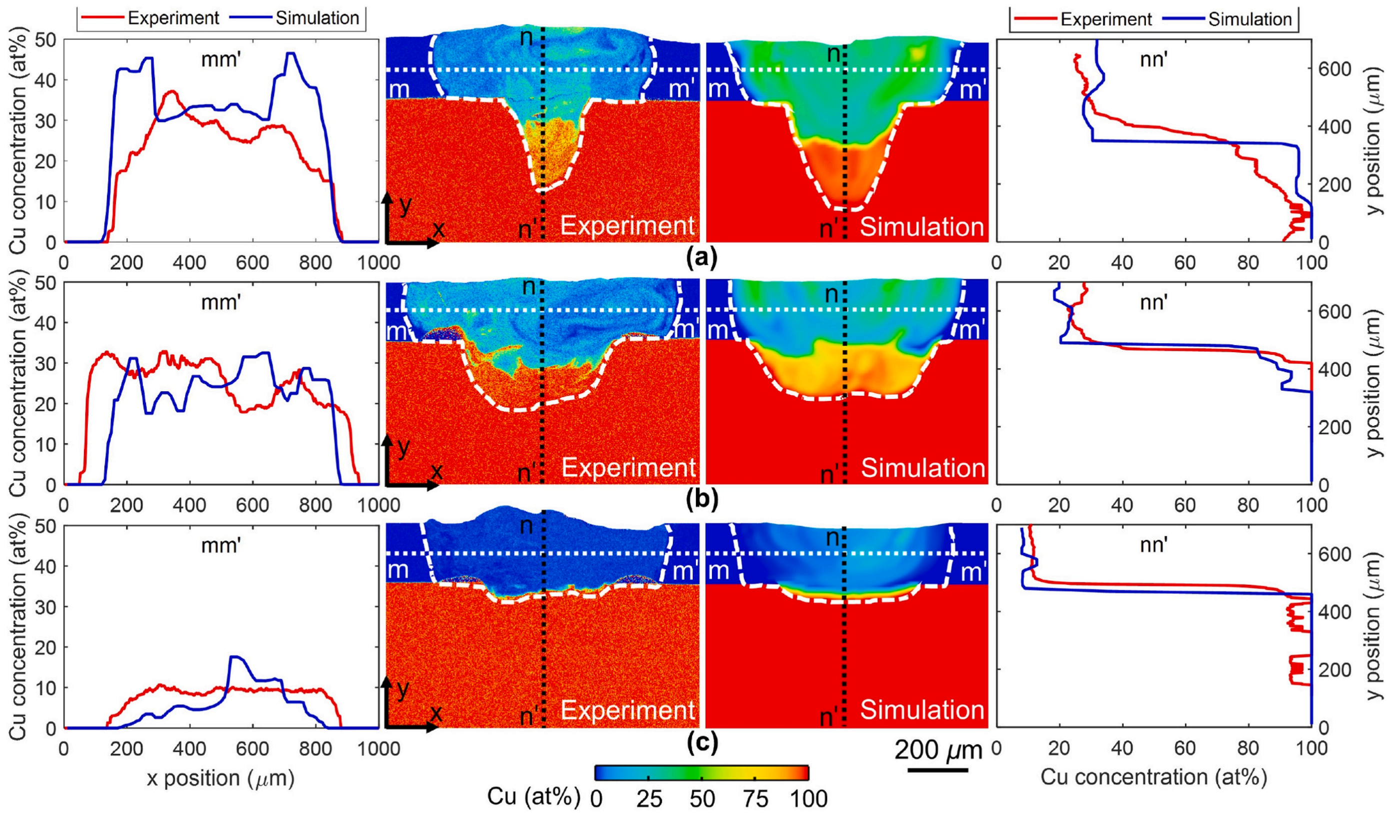

Fig. 7. Comparison of the experimental and simulation results of the fusion zone geometry and Cu concentration field for (a) case #1, (b) case #13, and (c) case #16.

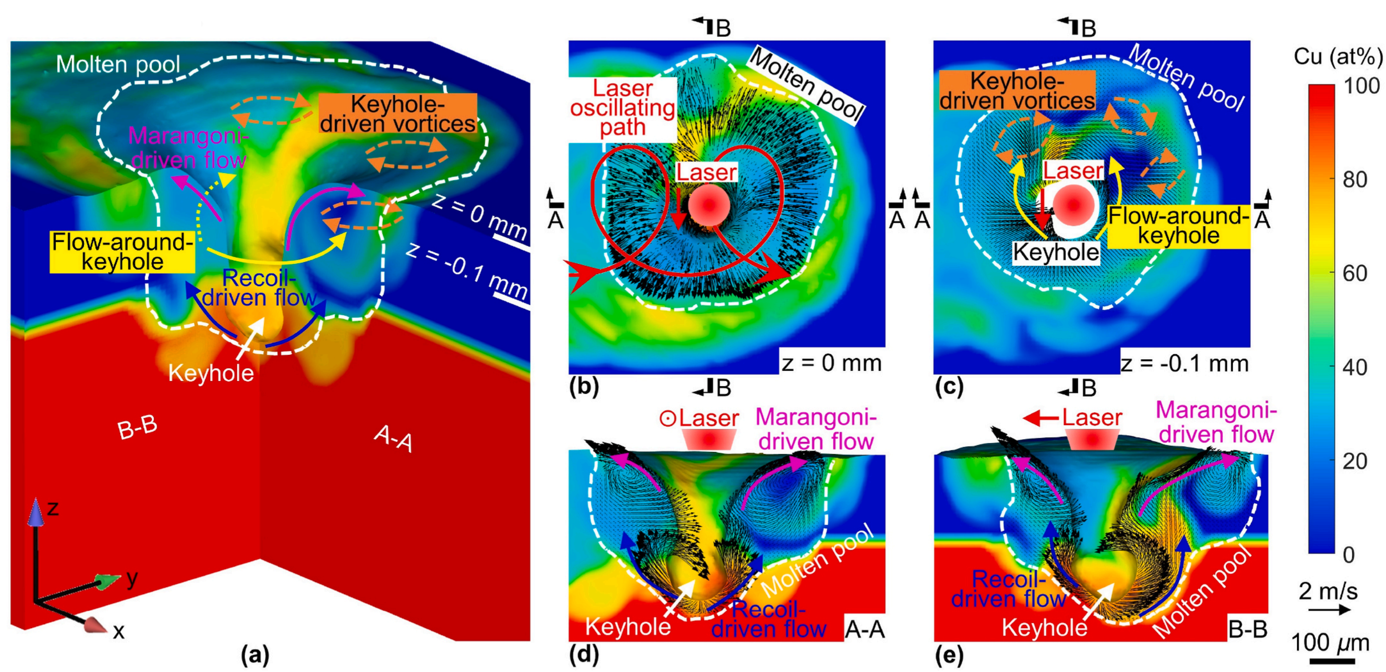

Fig. 8. (a) A 3D snapshot of the Cu concentration field (color map) and fluid flows (color arrows) in the molten pool at 13.25 ms of the simulation result for case #9. Cu concentration field (color map), flow velocity field (black arrows), and fluid flows (color arrows) of (b) top-view at z = 0 mm plane, (c) horizontal section view at z = −0.1 mm plane, (d) longitudinal section view at A-A plane, and (e) cross-section view at B–B plane. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

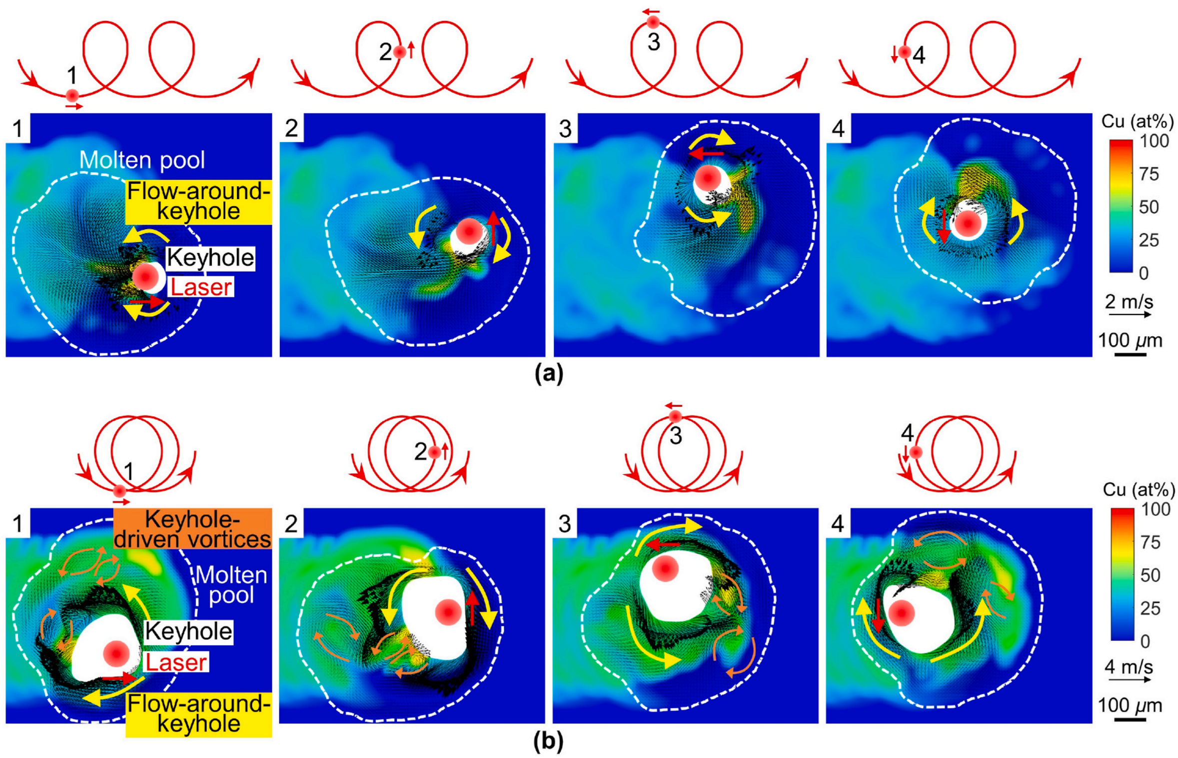

Fig. 9. Comparison of laser oscillating path (red curves), fluid velocity field (black arrows), flows (color arrows), and Cu concentration field (color map) in the molten pool for (a) case #9 and (b) case #12 at horizontal section z = −0.15 mm plane at four different locations of the laser oscillating path. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Reynolds number is between 40 and 200. Here the Reynolds number for the flow-around-keyhole can be calculated by

$$Re = \frac{\rho V_o d}{\mu} \tag{12}$$

where ρ is the density of the fluid, V_o is the keyhole moving speed which is the same as the laser oscillating speed defined in Eq. (1), d is the keyhole diameter, and μ is the fluid dynamic viscosity. The Reynolds number is 25 for case #9 and 128 for case #12, which, according to the flow-around-a-circular-cylinder analysis, suggests vortices in case #12 but not case #9. This is consistent with the CFD simulation results.

Fig. 10 presents a series of data to quantitatively investigate the effects of oscillating amplitude and frequency on the four fluid flows and metal mixing. The effects of oscillating amplitude on flow velocities and metal mixing are shown in column (a) of Fig. 10. For cases #1, #5, #9, and #13, the oscillating frequency is fixed at 300 Hz, while the oscillating amplitude increases from 0.05 mm to 0.20 mm. Therefore, the laser oscillating speed increases linearly from 0.094 m/s to 0.377 m/s (red curves in column (a) of Fig. 10) according to Eq. (1).

The trends for recoil-driven flow and Marangoni-driven flow are consistent with the findings in a previous work [9]. The maximum recoil-driven flow velocity decreases from 10.31 m/s to 6.85 m/s (blue curves in column (a) of Fig. 10). While the maximum recoil pressures on the keyhole wall are similar in the four cases (~1.45×10^6 Pa) due to the identical laser power, the increased laser oscillating speed allows less time for the fluid in certain areas to be accelerated by the recoil pressure. The maximum recoil-driven flow velocity is, therefore, decreased. The maximum Marangoni-driven flow velocities remain similar among the four cases (~2.50 m/s, magenta curves in column (a) of Fig. 10) because the identical laser power leads to very similar Marangoni stress (~8500 N/m^2).

The maximum velocity of the flow-around-keyhole increases from 6.94 m/s to 9.51 m/s (yellow curves in column (a) of Fig. 10), which is directly driven by the fast-moving keyhole with an increasing oscillating speed. The Reynolds number for the flow-around-keyhole is increased from 9 to 33 based on Eq. (12) (orange curves in column (a) of Fig. 10), which is primarily due to the increment of keyhole moving speed and keyhole diameter, both of which are increased with an increasing laser oscillating speed. The Reynolds number is also dependent on the density and viscosity of molten metal, both of which are dependent upon the mixing ratio of the two dissimilar metals. In these four cases, the density and viscosity are relatively consistent, and do not significantly change the Reynolds number. The threshold Reynolds number for keyhole-driven vortices formation is 40 according to the flow-around-a-circular-cylinder analyses, and is marked out by the black dashed line. It suggests that no keyhole-driven vortices will be formed in the molten pool for the four cases, which is consistent with the CFD simulation results.

Due to the decreasing recoil-driven flow velocity and the molten pool depth, less Cu is driven to flow upward from Region II to mix with the Al in Region I. Therefore, $\mu^{\mathrm{I}}_{\mathrm{Cu}}$ is decreased from 34.16 at% to 26.89 at% for the four cases, as the oscillation amplitude is increased from 0.05 mm to 0.20 mm (solid green curves in column (a) of Fig. 10), which matches well with the experimental measurements (dashed green curves in column (a) of Fig. 10). In the meantime, $\sigma^{\mathrm{I}}_{\mathrm{Cu}}$ only shows minor variation in both the simulation results (solid cyan curves in column (a) of Fig. 10) and experimental results (dashed cyan curves in column (a) of Fig. 10), indicating that the uniformity of Al–Cu mixing does not change significantly in the four cases. This is the compromise between two conflicting effects: on one hand, the increasing velocity of flow-around-keyhole can slightly improve the Al–Cu mixing; on the other hand, an increasing laser oscillating speed causes a decreasing molten pool lifetime from 8.70 ms to 5.13 ms (purple curves in column (a) of Fig. 10), which allows less time for the Al–Cu mixing.

The effects of oscillating frequency on flow velocities and metal mixing are shown in column (b) of Fig. 10. For cases #13, #14, #15, and #16, the oscillating amplitude is fixed at 0.20 mm, while the oscillating

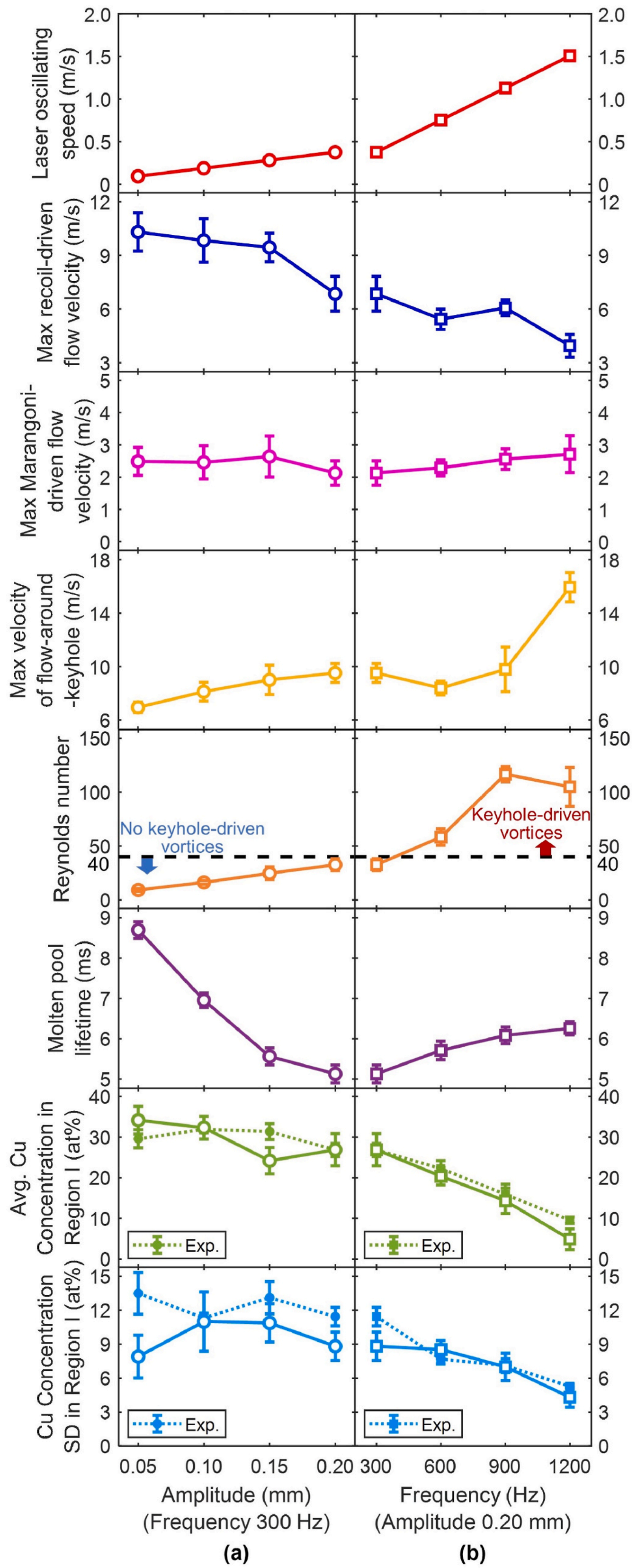

Fig. 10. Effects of oscillating amplitude and frequency on laser oscillating speed, maximum recoil-driven flow velocity, maximum Marangoni-driven flow velocity, maximum velocity of flow-around-keyhole, Reynolds number, molten pool lifetime, average Cu concentration in Region I ($\mu^{\mathrm{I}}_{\mathrm{Cu}}$), and Cu concentration standard deviation in Region I ($\sigma^{\mathrm{I}}_{\mathrm{Cu}}$) based on simulation results. The error bars are 1 standard deviation and represent the variations in the value of the y-axis.

frequency is increased from 300 Hz to 1200 Hz. According to Eq. (1), the laser oscillating speed increases linearly from 0.377 m/s to 1.508 m/s (red curves in column (b) of Fig. 10).

The maximum recoil-driven flow velocity decreases from 6.85 m/s to 3.95 m/s (blue curves in column (b) of Fig. 10), the maximum Marangoni-driven flow velocities are ~2.50 m/s (magenta curves in column (b) of Fig. 10), the maximum velocity of the flow-around-keyhole increases from 9.51 m/s to 15.95 m/s (yellow curves in column (b) of Fig. 10), and the Reynolds number increases from 33 to 105 (orange curves in column (b) of Fig. 10). These four trends are similar to the effects of increasing oscillating amplitude on the four metrics, and can be explained by the same physical reasons. Moreover, the Reynolds numbers of cases #14 - #16 are above the threshold value, which suggests that keyhole-driven vortices will be formed in the molten pools in these cases, which is consistent with the CFD simulation results. It is interesting to note that the Reynolds number does not increase monotonously but drops from 117 in case #15 to 105 in case #16. This is because the Reynolds number is also dependent on the molten pool density which is decreased from case #15 to case #16 due to a decreasing $\mu^{\mathrm{I}}_{\mathrm{Cu}}$ (solid green curves in column (b) of Fig. 10).

As the recoil-driven flow velocity and the molten pool depth both decrease, less Cu is driven to flow upward from Region II to Region I, therefore $\mu^{\mathrm{I}}_{\mathrm{Cu}}$ is decreased from 26.89 at% to 4.87 at% as the oscillation frequency increases from 300 Hz to 1200 Hz (solid green curves in column (b) of Fig. 10), which matches well with the experimental result (dashed green curves in column (b) of Fig. 10). Meanwhile, the Al–Cu mixing in the molten pool is improved, which is reflected by decreasing $\sigma^{\mathrm{I}}_{\mathrm{Cu}}$ values in both the simulations (solid cyan curves in column (b) of Fig. 10) and experiments (dashed green curves in column (b) of Fig. 10). This decreasing trend is due to three physics: first, the laser with a higher oscillating frequency will revisit the tail of the molten pool more frequently, and the four fluid flows will mix the local metals more times to improve its uniformity; second, the formation of keyhole-driven vortices can also facilitate the local metal mixing; third, the laser beam frequently circles back to revisit the fusion zone/molten pool, which lengthens the molten pool lifetime from 5.13 ms to 6.26 ms (purple curves in column (b) of Fig. 10), which allows more time for the Al–Cu mixing.

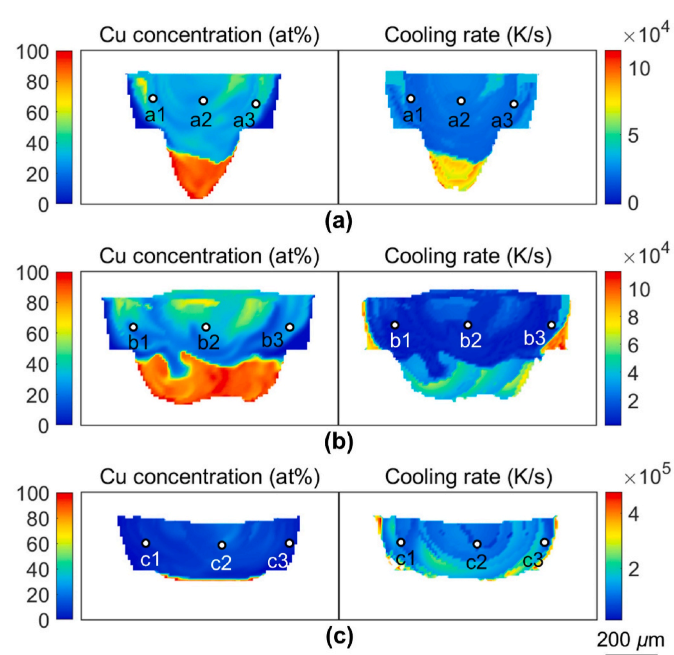

Fig. 11. Simulated Cu concentration and cooling rate fields (thermo-solutal conditions) of (a) case #1, (b) case #13, and (c) case #16 from the CFD model.

3.3. Computational materials model

The microstructure of the weld is determined by the local thermo-solutal condition during welding. The thermo-solutal condition is difficult to obtain from experiments, but can be predicted by the CFD model presented in the previous section. Shown in Fig. 11 are the predicted thermo-solutal conditions for the three cases in Fig. 7 (cases #1, #13, and #16). It can be found that the distributions of the cooling rate are asymmetric. This is because the advancing and retreating sides of the laser oscillating path (as labeled in Fig. 1) will generate different conditions for the laser absorption, heat transfer, fluid flow, and metal mixing, all of which will lead to different cooling rates on the two sides. It is also interesting to note that the cooling rate field shows a ring-shaped pattern in case #16. This is because the laser of a high oscillating frequency revisits the same location multiple times, which causes the fusion zone to re-melt and re-solidify layer-by-layer, generating the ring-shaped pattern of the cooling rate field.

Different locations of the fusion zone have different thermo-solutal conditions, which result in different local microstructures. Three spots are selected from the left, center, and right portions of Region I from the simulation results (marked as white dots in Fig. 11), and the predicted thermo-solutal conditions are used in the computational materials model to simulate the phase formation at those spots. The simulation result of the solidification path for spot b1 is presented here to demonstrate the result, as shown in Fig. 12. With the Cu concentration being 27.54 at% and the cooling rate being 4462 K/s for spot b1, primary θ first precipitates from the liquid molten pool when the temperature reaches 862.3 K and grows to 66.5 vol% as the temperature drops to the eutectic temperature of 820.7 K. Then all the remaining liquid molten pool transforms into the eutectic phase of 33.5 vol%, with 16.8 vol% eutectic θ phase and 16.7 vol% of eutectic α phase. The corresponding experimental result of the microstructure for spot b1 was taken by SEM backscattered electrons imaging at the same locations of the fusion zone, as shown the case #13 in Fig. 5. The simulation result matches reasonably well with the experimental result, which, based on the image processing result, is composed of 60.5 vol% of primary θ and 39.5 vol% of eutectic.

The model has been applied to simulate all the spots listed in Fig. 11. Their thermo-solutal conditions and predicted phase compositions are listed in Table 3. The corresponding experimental results of Cu concentration and phase composition for all the spots are obtained at the same locations of the fusion zone from the welding experiments and are listed in Table 3. The predicted volume percentages of primary α, eutectic, and primary θ at all the spots agree reasonably well with the experimental results.

The model has been further applied to simulate the phase distribution in the entire fusion zone for different cases, and the predicted phase distribution for cases #1, #13, and #16 are shown in Fig. 13. In case #1, μ^{I}_{Cu} is very close to the composition of the θ phase (as shown in Fig. 7(a)), and the resultant microstructure is dominated by primary θ with only a small amount of eutectic phase dispersed therein (as shown in Fig. 13(a)). With such a high-volume percentage, primary θ will likely exist in the bulk format and present a high brittleness, which will lead to a high crack tendency for the weld. In case #13, μ^{I}_{Cu} is reduced to the hyper-eutectic region (as shown in Fig. 7(b)), so Region I is almost equally shared by the primary θ and eutectic phase (as shown in Fig. 13(b)). Based on the predicted solidification path by the model, the most likely morphology of the microstructure will be primary θ granules surrounded by a eutectic network. The size of the primary θ granules cannot be directly predicted by the model, but should decrease with an increasing cooling rate. This microstructure should provide reasonably good mechanical strength. In case #16, μ^{I}_{Cu} is further reduced to the hypoeutectic region (as shown in Fig. 7(c)), and Region I is occupied by the primary α and eutectic phase (as shown in Fig. 13(c)). The microstructure morphology should be very similar to that of case #13, except that the primary θ is replaced by primary α.

The process-microstructure-property relationship of oscillating laser welding of Al–Cu can be summarized as follow: oscillating laser beam can significantly change the fluid flow pattern and metal mixing in the molten pool. With the oscillating parameter selected appropriately, the chemical composition and its homogeneity in the welds can be effectively controlled. If the final chemical composition falls into the hypereutectic regime of the Al–Cu phase diagram, a microstructure morphology of granular-shaped IMC surrounded by eutectic phases can be formed in the welds. The continuous eutectic network provides a strong and ductile backbone and the IMC phase acts as an effective reinforcement, and thus the welds can sustain relatively high mechanical loadings.

4. Conclusions

The objective of this work is to reveal the fundamental physics that governs the fluid flow, heat/solute transfer, and microstructure evolution in the molten pool in oscillating laser welding of dissimilar metals. To achieve this, a combination of experiments and simulations was used in this work to investigate oscillating laser welding of Al–Cu lap joints. Parametric experiments were conducted to identify the effects of oscillating amplitude and frequency on the final metal distribution, phase composition and morphology, and mechanical performance of the welds. A simulation framework was built to understand the fundamental physics: a CFD model was developed to reveal the fluid flow and metal mixing processes during welding; a Scheil solidification model was developed to predict the phase distributions in the welds based on the predicted thermo-solutal conditions from the CFD model. The major findings of the work are summarized as follows.

1. The metal mixing in the top region of the molten pool is driven by four different fluid flows, namely recoil-driven flow, Marangoni-

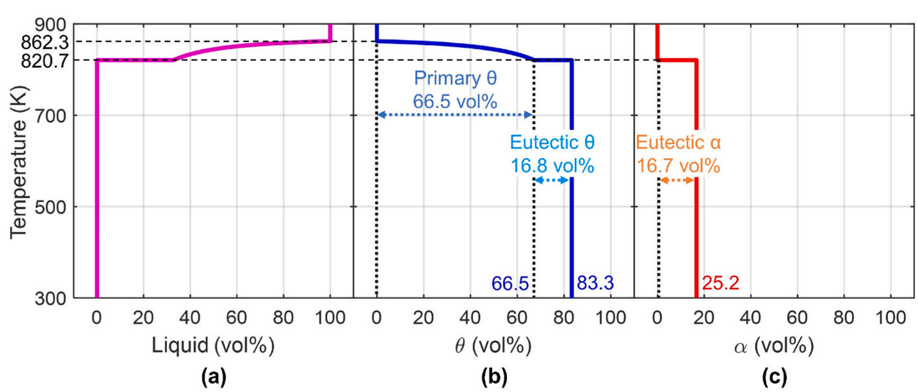

Fig. 12. Predicted solidification path of spot b1 in Fig. 11.

Table 3
Comparison of the experimental and simulation results of the phase formation of the selected spots in Fig. 11.

Data source		Cu (at%)		Cooling rate (K/s)	Primary α (vol%)		Eutectic (vol%)		Primary θ (vol%)	
Exp.	Sim.	Exp.	Sim.	Sim.	Exp.	Sim.	Exp.	Sim.	Exp.	Sim.
Left	a1	31.77	32.09	16,500	0	0	4.5	5.6	95.5	94.4
Center	a2	27.46	27.14	21,030	0	0	31.7	36.1	68.3	63.9
Right	a3	25.30	24.99	15,570	0	0	44.8	49.4	55.2	50.6
Left	b1	27.57	27.54	4462	0	0	39.5	33.5	60.5	66.5
Center	b2	30.74	30.19	7147	0	0	10.3	16.0	89.7	84.0
Right	b3	28.86	28.23	10,610	0	0	26.6	28.9	73.4	71.1
Left	c1	6.17	5.85	87,610	70.4	71.3	29.6	28.7	0	0
Center	c2	7.92	8.27	113,700	60.1	57.8	39.9	42.2	0	0
Right	c3	4.49	4.68	63,400	77.5	77.6	22.5	22.4	0	0

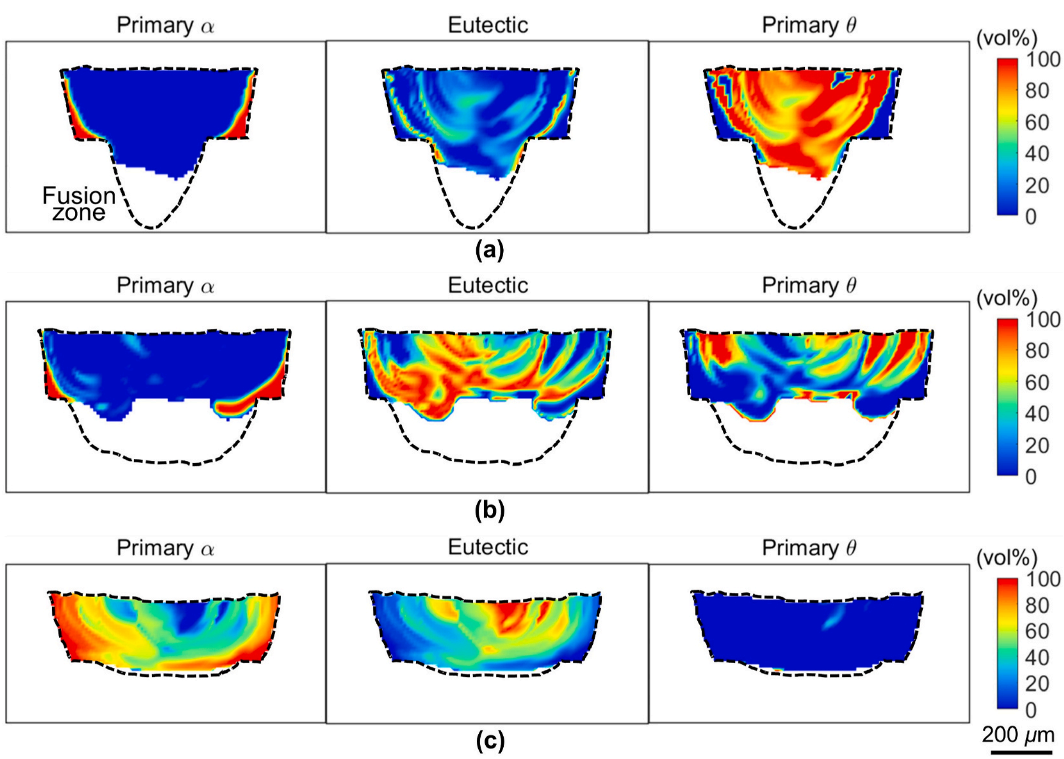

Fig. 13. Simulated phase distribution fields in the fusion zone for cases #1, #13, and #16 listed in Fig. 11.

driven flow, flow-around-keyhole, and keyhole-driven vortices, which are influenced by laser oscillating parameters. More specifically,

- As the oscillating amplitude and frequency increase, the recoil-driven flow velocity and molten pool depth decrease, reducing the amount of Cu that flows upward and hence reducing the Cu concentration in the top region of the molten pool.
- An increasing oscillating amplitude or frequency does not alter the Marangoni-driven flow but increases the flow-around-keyhole velocity. As the Reynolds number for the flow-around-keyhole increases, keyhole-driven vortices will form to improve the uniformity of Al–Cu mixing.
- A higher oscillation frequency will revisit the tail of the molten pool more frequently, allowing the four fluid flows to mix the local metals more repeatedly with more uniform mixing.
- A decreasing oscillating amplitude or increasing oscillating frequency will lengthen the molten pool lifetime, which allows the four fluid flows to have longer interactions and mix Al–Cu more uniformly.

2. The fusion zone microstructure is greatly affected by the dissimilar metal mixing in the fusion zone. As the average Cu concentration in the top fusion zone region gradually increases from the hypoeutectic to eutectic and then hypereutectic, the dominating microstructure in the welds gradually changes from granular primary α to eutectic phase, to granular primary θ, and then to bulk primary θ.
3. The weld peak load highly depends on the phase composition and morphology in the top region of the fusion zone. The microstructure of primary θ or α granules surrounded by the eutectic phase with fine lamellae provides good mechanical strength for the welds; the microstructure dominated by bulk primary θ will cause the welds to be susceptible to crack formation and propagation.

This work provides a better understanding of the complex phenomena in oscillating laser welding of Al–Cu from a combined experimental and simulation perspective. In addition, this work has revealed the effects of laser oscillation parameters on fluid flow, metal mixing, microstructure formation, and mechanical performance of the welds, all providing valuable information for the process optimization toward reliable welds of dissimilar metals. It is recommended to use a higher oscillating frequency and amplitude to achieve a homogeneously mixed fusion zone and control the average Cu concentration in the fusion zone below 33.3 at% by tuning the fusion zone depth with the adjustments of laser power and laser forward speed.

While it can be extremely difficult to eliminate IMC formation in certain welding situations, it is possible to manipulate the chemical composition of the molten pool so that the final microstructure will not be negatively affected by the IMC phases. For many dissimilar metal pairs that form IMCs, it is common to find in their phase diagrams a eutectic reaction between one solid solution phase and one IMC phase. If the molten pool composition can be restrained at or below the hypereutectic levels, the IMC phase will be a part of or surrounded by the eutectic structure in the final microstructure (such as in cases #9 - #16 of this work). This is beneficial because the high hardness of the IMC phase can contribute to the high strength of the microstructure while its high brittleness can be remedied by the ductile solid solution phase in the eutectic structure. By adjusting the laser power and forward speed, one can tune the molten pool shape (i.e., the volume of the melted metals) and hence control the molten pool composition at or below the hypereutectic levels. And by applying proper laser oscillating parameters (pattern, amplitude, and frequency), one can ensure a homogeneous distribution of two elements in the molten pool, avoiding certain regions having particularly high compositions and hence the formation of IMC phase(s) in the bulk form. With appropriate selections of all the processing parameters, oscillating laser welding can produce welds with desired chemical composition, microstructure, and performance. This strategy should work beyond laser and be applicable to other heat sources (e.g., electric arc and electron beam) in fusion-based dissimilar metal joining and multi-material additive manufacturing processes.

Credit author statement

Wenkang Huang: software, validation, investigation, formal analysis, writing - original draft. Wayne Cai: conceptualization, resources, writing – review & editing. Teresa Tinker: resources, investigation, writing – review & editing. Jennifer Bracey: supervision, writing – review & editing. Wenda Tan: conceptualization, project administration, writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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